Yonghui Liu

Ph.D. Candidate, KAIST | Former Software Engineer, Meituan

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Spatiotemporal AI | **Generative Modeling** | **Representation Learning**

Research spans motion prediction and reconstruction, transferable representations, and controllable traffic generation. Published machine learning studies and current simulation and sensing prototypes connect research methods with system development.

EDUCATION

KAIST Aug 2023 - Present
Ph.D. candidate, Cho Chun Shik Graduate School of Mobility; advisor: Prof. Inhi Kim

Zhejiang University Sep 2018 - Mar 2021
M.S. in Ocean Engineering

Nanchang University Sep 2014 - May 2018
B.S. in Physics

PROFESSIONAL EXPERIENCE

Software Engineer | Meituan May 2021 - Aug 2023

AI Research Intern Apr 2020 - Oct 2020
Advanced Institute of Information Technology, Peking University

PUBLISHED RESEARCH

- **Trajectory reconstruction:** Developed ProChunkFormer, a progressive chunked Transformer for recovering road-consistent trajectories from sparse and noisy GPS.
- **Motion prediction analysis:** Proposed SDA, a model-agnostic protocol attributing forecasting performance to historical motion, current state, and goal information.
- **Transferable representations:** Combined LLM-derived semantics with heterogeneous graph learning for road embeddings and cross-city generalization.
- **Low-supervision learning:** Combined LLM pseudo-labels and structural embeddings for cross-taxonomy alignment; evaluated with as little as 1% human-labeled data.

CURRENT RESEARCH PROJECTS

Route-Conditioned Traffic Simulation | Ongoing research

Developing route-conditioned traffic generation and joint fine-tuning on SMART/CAT-K. Implemented recorded-scene adapters and offline intent-aware route allocation under feasibility and quota constraints; closed-loop evaluation is ongoing.

Autonomous Traffic Sensor Web (ATSW) | Working research prototype

Built a local GPS replay web workspace with heuristic adaptive reporting, Kalman receiver-state estimation, historical reconstruction, playback controls, decision logs, and JSON run export.

SELECTED PUBLICATIONS

1. **Y. Liu** and I. Kim. Sequential decomposition and attribution for trajectory forecasting. **Knowledge-Based Systems**, 335:115204, **2026**.
2. **Y. Liu**, Q. Li, and I. Kim. Enhanced trajectory reconstruction from sparse and noisy GPS data: A progressive chunked transformer approach. **Communications in Transportation Research**, 5:100200, **2025**.
3. **Y. Liu** and I. Kim. A Generalizable Semantic Graph Learning Framework for Road Network Representation. **IEEE ITSC**, pp. 720-725, **2025**.
4. **Y. Liu** and I. Kim. Cross-Taxonomy Label Alignment in Transportation Networks With Prompt-Driven Large Language Models. **IEEE ITSC**, pp. 1724-1729, **2025**.

Additional co-authored journal articles: **Energy** (2022); **Journal of Geophysical Research: Earth Surface** (2021). Full publication list online.

TECHNICAL SKILLS

Methods: Transformers, GNNs, LLM-based semantic supervision, conditional generation, Kalman filtering.
Tools: Python, PyTorch, FastAPI, React, TypeScript, SQLite.